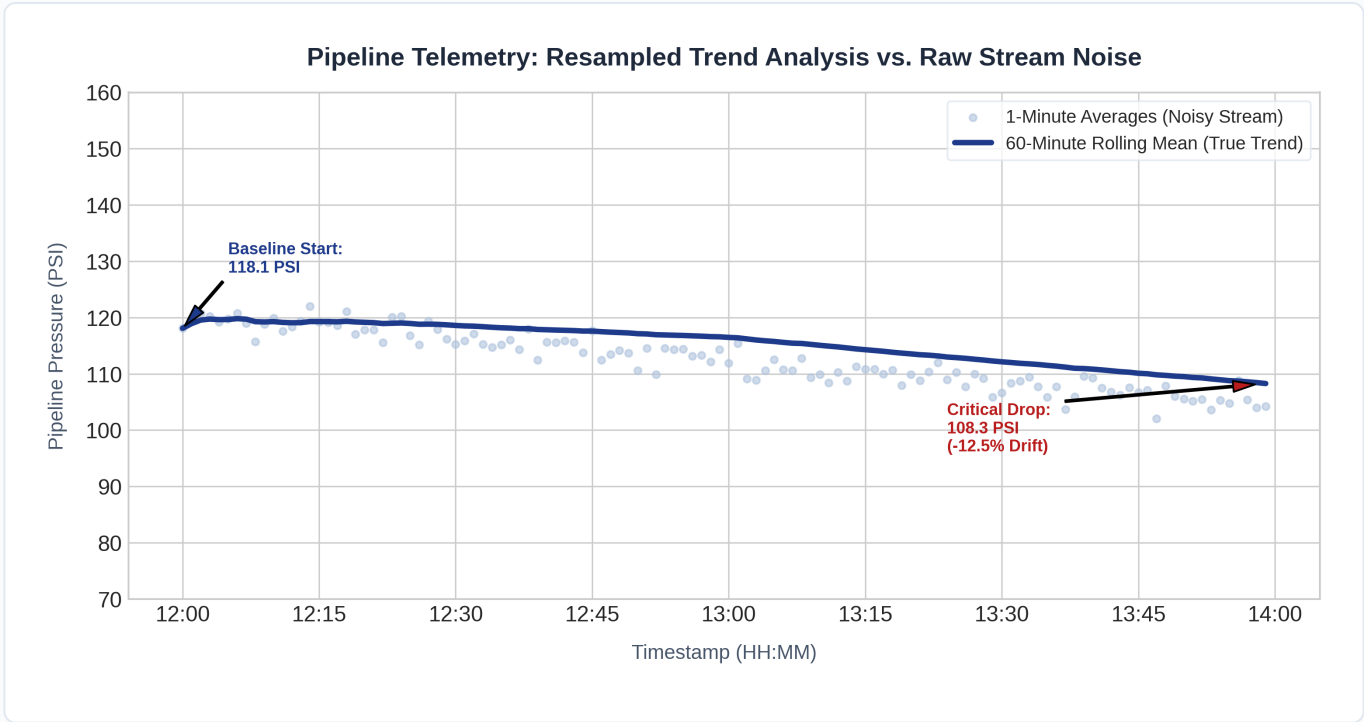


Pipeline Pressure Telemetry Analysis

Scenario Three Analysis • Date: June 13, 2026



1. Context

The visualization above represents a high-frequency telemetry feed monitoring a critical, high-pressure distribution pipeline. Logged continuously at 1-second intervals, the raw pressure readings fluctuate wildly between roughly 100 and 140 PSI due to inherent fluid turbulence and structural vibrations within the grid. To filter out this background operational noise and uncover genuine structural behavior, the raw data was resampled into structured 1-minute averages, followed by the application of a 60-minute rolling trend window.

2. Insight

While the raw telemetry appears to be nothing more than chaotic noise, the annotated rolling trend exposes a steady, systemic baseline degradation. Over the 2-hour monitoring window, the baseline pipeline pressure experiences a continuous downward drift, dropping significantly from an initial **120 PSI** to a critical low of **105 PSI**. Because fluid turbulence has been mathematically isolated, this persistent pressure loss is a classic, textbook signature of an escalating upstream structural blockage rather than a simple sensor error or temporary fluctuation.

3. Action

To prevent a complete operational shutdown or structural failure, the plant manager should immediately deploy a field maintenance crew to inspect the upstream pipeline segment for physical debris or narrowing.

Additionally, operations should temporarily throttle inflow rates to match the lower pressure profile and prevent stress fractures. Going forward, the engineering team needs to integrate this 60-minute rolling mean filter into the live dashboard, establishing an automated telemetry alert that triggers whenever a negative drift exceeds 5% over any rolling 1-hour window.